Ketan Sharma

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OBJECTIVE

College student seeking an internship in the field of mechanical design in Silicon Valley.

EDUCATION

Bachelor of Science (B.S.) in Mechanical Engineering May 2028
San Jose State University; San Jose, CA GPA: 3.42
Relevant Coursework: Mechatronics, Machine Operations & Safety, Machining Processes

SKILLS

Design: SolidWorks, GD&T, DFM

Fabrication: 3D Printing, Machining, Woodworking, Sheet Metal

Electronics & Programming: Raspberry Pi Pico, Arduino, Python, C++

PROJECTS

Automated Medication Dispenser

- Designed and 3D printed a seven-compartment rotating carousel mechanism to house and organize individual medication pills for timed doses.
- Integrated a microcontroller, wired electrical components, and multi-sensor feedback logic with a user interface to automate weekly medication dispensing, verify pill delivery, and monitor user medication retrieval.
- Achieved 90% accuracy (18/20) in pill-drop detection through RTC-driven scheduled dispensing control.

Wall Mounted Hanger

- Machined a two-part, heavy-duty aluminum wall-mount hanger: a milled plate and a lathed peg.
- Applied manufacturing processes to meet precision fit requirements, including countersinking, threading, and assembling a #10-24 flat machine screw between the lathe and mill parts.
- Achieved finished product dimensions within ±0.005" of target tolerances.

Mini Mailbox

- Fabricated a miniature hinged door mailbox through sheet metal and woodworking processes.
- Assembled mailbox with a functioning hinged door, holding almost all fit tolerances within ±1/16" of the drawing specifications.

Thermoelectric Generator Senior Project, Branham High: https://tinyurl.com/thermoelectricgen

- Designed and tested thermoelectric wearable wrist prototype, validating the concept of heat-to-electricity conversion.
- Defined design criteria and implemented test framework to assess prototype performance and materials.
- Collected and analyzed survey data from students to identify user needs, constraints, parameters, and design specifications.

WORK EXPERIENCE

Technology Consultant, Target, San Jose West, CA September 2023 – August 2025

- Delivered personalized technical solutions through assessing customer demands and expectations.
- Resolved and clarified product features, software and hardware issues, enhancing customer understanding and improving overall satisfaction.
- Monitored inventory and pricing strategies to optimize sales performance through floor products and placements.